

A Comprehensive Proof of Bertrand's Theorem*

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Abstract. A cornerstone result in Newtonian mechanics is Bertrand's Theorem concerning the behavior of the solutions of the classical two-body problem. It states that among all possible gravitational laws there are only two exhibiting the property that all bounded orbits are closed. One of these is Newtonian gravitation, the other being Hookean gravitation. We present a comprehensive proof of Bertrand's Theorem that is accessible to undergraduate students who are familiar with basic notions from advanced calculus and differential equations.

Key words. classical mechanics, Bertrand's Theorem, Newtonian gravitation, Hooke's law

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I. Introduction. High school students taking a first course in physics or mechanics often learn that Newton's law of gravitation states that the mutual force of attraction between two point masses m_1 and m_2 is proportional to both masses, as well as proportional to the reciprocal of the square of the distance r between the masses. That the magnitude of the force depends on the distance between the masses seems reasonable when one accepts the isotropic nature of space. But why is the universe governed by a power law that is proportional to r^{-2} ? What would happen if the law of gravity did not take the form of a power law? Or what if it were a power law, but with a power that isn't equal to -2 ?

When postulating Newtonian gravity, one can prove the laws of Kepler, the first of which states that the orbit of one of the masses always traces out a curve with a specific shape around the other mass, namely, a conic section (ellipse, parabola, or hyperbola). Bounded orbits, in particular, must be ellipses, and obviously these are closed curves. Furthermore, various geometrical and dynamical features of these orbits can be derived explicitly, which consolidates Kepler's second and third laws. To this day, this accomplishment remains one of the major feats of the scientific thought process because it demystifies the painstakingly observed patterns in the motion of

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celestial objects that people had been noticing throughout the history of humankind. But it does not resolve the aforementioned questions.

Newton was aware that there are in fact two specific gravitational laws having the property that all bounded orbits are closed; see Propositions X and XI in Book 1 in [6]. Both are power laws. One of them is, of course, Newtonian gravitation (this follows from Kepler's first law). The other one is obtained if gravitation is governed by Hooke's law, which postulates that the force of attraction is directly proportional to the distance r between the masses; this case corresponds to the force exerted by a linear spring connecting both masses. However, Newton did not report whether there were other gravitational laws with this property. It took about two more centuries before Bertrand announced in 1873 in [2] that there are none. Since then, this result has been known as Bertrand's Theorem. Note that of these two laws, Newtonian gravitation is the only one that decays to zero as the distance r goes to infinity. That we live in a universe governed by the only gravitational law having these seemingly reasonable properties is nothing short of remarkable.

The implications of Bertrand's Theorem are so profound that it has been integrated into the undergraduate curriculum in theoretical mechanics; see, for instance, [3]. But to the best of our knowledge, there is still no comprehensive proof of Bertrand's Theorem available in the literature that is accessible to undergraduate mathematics or physics students. While Bertrand's original paper [2] (see also [7] for a recent English translation) is concise and includes some key ideas, it does not contain what would nowadays be considered to be a proof of Bertrand's Theorem. On pages 36–38 in [1], Arnold laid out a sketch of a proof by stating a list of six problems. Although he provided short, one line answers to each of these problems, he did not include proofs. Moreover, there are several typos in both the statements of and corresponding answers to these problems. Furthermore, only relatively recently did Jovanović in [4] point out that “details lack and it turns out that they are not so easy to prove.” In fact, the purpose of [4] is precisely to give a proof for one of Arnold's problems, namely, Problem 5 in [1]. Although a solution to Problem 6 in [1] can initially proceed along the same lines as the solution to Problem 5 presented by Jovanović, we believe that additional complications arise that require further attention and new ideas. One of the goals of this paper is therefore to provide a solution to Problem 6 in [1]. Also, due to the fact that a comprehensive proof of Bertrand's Theorem is not readily available, it is the main goal of this paper to provide one.

This work represents research carried out by two undergraduate Mathematics students in collaboration with a faculty advisor. As such, parts or all of this work can serve as a project for undergraduate students in mathematics or physics who have taken the classical calculus sequence courses and a first course in differential equations, and who are familiar with some vector calculus and certain concepts from advanced calculus.

In the rest of this paper we follow the usual convention to denote vector-valued quantities by boldface letters.

2. The Two-Body Problem. Let $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ denote the location in $\mathbb{R}^3$ of two constant point masses m_1 and m_2 at time $t \geq 0$. We assume that for all t both masses exert opposite, attractive forces upon each other which are directed along the vector $\mathbf{r}(t) := \mathbf{r}_2(t) - \mathbf{r}_1(t)$ that connects them, and that the magnitude of this force depends only on the distance $r(t) := \|\mathbf{r}(t)\|$ between the masses. Furthermore, we assume that the magnitude of the force is modeled by a smooth, positive function $h : (0, +\infty) \rightarrow (0, +\infty)$. For example, Newtonian gravitation corresponds to $h(r) = Gm_1m_2/r^2$,

where $G = 6.67 \times 10^{-11} Nm^2/kg^2$ is the gravitation constant, and Hookean gravitation corresponds to $h(r) = kr$, where k is a positive constant known as the spring constant, because we can interpret this type of gravitation as being caused by a linear spring connecting both masses.

We shall use $\dot{\mathbf{r}}_1(t)$ and $\dot{\mathbf{r}}_2(t)$ to denote the respective velocities $d/dt(\mathbf{r}_1(t))$ and $d/dt(\mathbf{r}_2(t))$ of the masses m_1 and m_2 at time t , and $\ddot{\mathbf{r}}_1(t)$ and $\ddot{\mathbf{r}}_2(t)$ for their accelerations.

Applying Newton's law of motion to each of the masses yields

$$(1) \quad m_1 \ddot{\mathbf{r}}_1(t) = h(r(t)) \frac{\mathbf{r}(t)}{r(t)},$$

$$(2) \quad m_2 \ddot{\mathbf{r}}_2(t) = -h(r(t)) \frac{\mathbf{r}(t)}{r(t)}.$$

The two-body problem amounts to determining the solutions $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ of this system of differential equations for all $t > 0$, given the initial locations $\mathbf{r}_1(0)$ and $\mathbf{r}_2(0)$ and the initial velocities $\dot{\mathbf{r}}_1(0)$ and $\dot{\mathbf{r}}_2(0)$ of the masses at time $t = 0$.

It is well known that this problem can be simplified by converting it into the problem of determining the location of a single, fictitious point mass in a central force field. We shall briefly review this next.

Let $\mathbf{R}(t) = (m_1 \mathbf{r}_1(t) + m_2 \mathbf{r}_2(t)) / (m_1 + m_2)$ be the location of the center of mass of m_1 and m_2 at time t , and define $m = (m_1^{-1} + m_2^{-1})^{-1}$, which is known as the reduced mass. Then switching from the unknown functions $(\mathbf{r}_1(t), \mathbf{r}_2(t))$ to $(\mathbf{R}(t), \mathbf{r}(t))$, (1)–(2) take the equivalent form

$$(3) \quad m \ddot{\mathbf{R}}(t) = 0,$$

$$(4) \quad m \ddot{\mathbf{r}}(t) = -h(r(t)) \frac{\mathbf{r}(t)}{r(t)}.$$

These equations are decoupled, and the first one is integrable,

$$\mathbf{R}(t) = \mathbf{R}(0) + \dot{\mathbf{R}}(0)t,$$

revealing that the center of mass of m_1 and m_2 obeys a uniform linear motion with respect to time t .

The second equation can be interpreted as an equation that describes the location $\mathbf{r}(t)$ in $\mathbb{R}^3$ at time t of a fictitious point mass m which is subject to an attractive central force with magnitude $h(r(t))$. It is well known that the motion of a point mass subject to a central force takes place in a fixed plane. Indeed, let

$$\mathbf{L}(t) = \mathbf{r}(t) \times m \dot{\mathbf{r}}(t)$$

denote the angular momentum of m at time t . Then

$$\dot{\mathbf{L}}(t) = \dot{\mathbf{r}}(t) \times m \dot{\mathbf{r}}(t) + \mathbf{r}(t) \times m \ddot{\mathbf{r}}(t) = 0 - \mathbf{r}(t) \times h(r(t)) \frac{\mathbf{r}(t)}{r(t)} = 0$$

for all t , implying that the angular momentum is conserved in time and equals $\mathbf{L}(0) = \mathbf{r}(0) \times m \dot{\mathbf{r}}(0)$. We assume throughout that $\mathbf{L}(0) \neq 0$. The conservation of angular momentum implies that $\mathbf{r}(t)$ and $\dot{\mathbf{r}}(t)$ must lie in a fixed plane, namely, in the plane spanned by the initial location and velocity vectors $\mathbf{r}(0)$ and $\dot{\mathbf{r}}(0)$. We can now focus

our attention on (4), but instead of assuming that $\mathbf{r}(t)$ belongs to $\mathbb{R}^3$, we may assume that it belongs to the above-mentioned plane. We say that the orbit of a solution $\mathbf{r}(t)$ of (4) is bounded if there is some $M > 0$ such that $\|\mathbf{r}(t)\| \leq M$ for all t , and we say that it is closed if $\mathbf{r}(t)$ is periodic, i.e., if there exists a nonzero T such that $\mathbf{r}(t+T) = \mathbf{r}(t)$ for all t .

Letting $(r(t), \theta(t))$ denote the polar coordinates of the location of the mass m in this plane, a somewhat tedious but straightforward calculation shows that (4) is equivalent to the following system of two scalar differential equations:

$$(5) \quad m(\ddot{r} - r(\dot{\theta})^2) = -h(r),$$

$$(6) \quad \frac{1}{r} \frac{d}{dt} (mr^2 \dot{\theta}) = 0,$$

where we have suppressed the explicit dependence on time t of the functions $r(t)$ and $\theta(t)$. Equation (6) can be integrated, and it amounts to the fact that the angular momentum is conserved,

$$(7) \quad m(r(t))^2 \dot{\theta}(t) = l_0$$

for all t . Here, l_0 is the signed magnitude of the initial angular momentum vector $\mathbf{L}(0)$. Note that the latter equation can be solved for $\dot{\theta}(t)$ in terms of $r(t)$ and then substituted into (5), yielding

$$(8) \quad m\ddot{r} - \frac{l_0^2}{m} \frac{1}{r^3} = -h(r).$$

Note that (7) implies that the angular coordinate $\theta(t)$ is a monotone function of time t . Whether $\theta(t)$ is increasing or decreasing depends only on the sign of l_0 . This suggests that we may as well consider the angular coordinate θ instead of time t as the independent variable. Furthermore, rather than treating r as the dependent variable as in (8), we shall consider its reciprocal

$$u = \frac{1}{r}$$

as the dependent variable. Using the chain rule, we can then rewrite (8) in the equivalent form

$$(9) \quad \frac{d^2 u}{d\theta^2} + u = f(u),$$

where $f : (0, +\infty) \rightarrow (0, +\infty)$ is a smooth function defined by

$$(10) \quad f(u) = \frac{m}{l_0^2} \frac{1}{u^2} h\left(\frac{1}{u}\right).$$

It is standard to convert this second-order differential equation into a system of two first-order equations by introducing a new variable $v(\theta) = du(\theta)/d\theta$,

$$(11) \quad u' = v,$$

$$(12) \quad v' = -u + f(u),$$

defined on the half-space $\mathcal{S} = \{(u, v) \in \mathbb{R}^2 \mid u > 0\}$. Here, the prime $'$ stands for $d/d\theta$. Although $\mathcal{S}$ may not be an invariant set for system (11)–(12), we shall study solutions

$(u(\theta), v(\theta))$ that remain in $\mathcal{S}$ for all θ . Such solutions will be shown to exist once we impose appropriate conditions on the gravitational law $h(r)$. Defining an energy-like function

$$E(u, v) = \frac{1}{2} (u^2 + v^2) - F(u),$$

where $F(u)$ is an antiderivative of the function f , i.e., $dF(u)/du = f(u)$, we see that along solutions $(u(\theta), v(\theta))$ of system (11)–(12) that remain in $\mathcal{S}$, the value of the function E remains constant:

$$\frac{dE(u(\theta), v(\theta))}{d\theta} = u(\theta)v(\theta) + v(\theta)(-u(\theta) + f(u(\theta))) - f(u(\theta))v(\theta) = 0.$$

For future reference we also introduce a function $G : (0, +\infty) \rightarrow \mathbb{R}$, defined by $G(u) = u^2/2 - F(u)$ for all $u > 0$. The functions $E(u, v)$ and $G(u)$ are therefore related as follows:

$$E(u, v) = \frac{1}{2} v^2 + G(u) \text{ for all } (u, v) \text{ in } \mathcal{S}.$$

To understand the behavior of the solutions of (4), we shall frequently switch between the equivalent systems (4), (5)–(6), and (11)–(12).

3. Preparing for Bertrand's Theorem.

3.1. Existence of Stable Circular Orbits. Bertrand's Theorem seeks to isolate those gravitational laws $h(r)$ for which all solutions $\mathbf{r}(t)$ of (4) with bounded orbits have closed orbits. In fact, and this is usually not mentioned explicitly in the literature, Bertrand's Theorem assumes a priori that (4) must exhibit *stable circular orbits* for all possible choices of the model parameters and initial conditions. More precisely, the following is assumed:

(SCO) $h(r)$ is such that for every positive value of m and every nonzero value of l_0 , (8) admits a constant solution $r(t) = r_0$ for some $r_0 > 0$, which is Lyapunov stable.

Equivalently, for every m and l_0 , system (11)–(12) must admit a Lyapunov stable steady state $(u_0, 0)$, where u_0 is a positive solution to the equation $f(u) = u$.

Remark 1. It should be noted that **(SCO)** does indeed restrict the class of targeted functions $h(r)$. Some choices for $h(r)$ do not even admit constant solutions $r(t) = r_0$ for (8) for all m and l_0 , let alone Lyapunov stable ones. For example, if

$$h(r) = \frac{1}{r^3} \left(\frac{1}{r} + 1 \right),$$

then (8) admits constant solutions $r(t) = r_0$ if and only if $l_0^2/m > 1$. Clearly, a necessary and sufficient condition for the existence of constant solutions for (8) is that the function $r \rightarrow r^3 h(r)$ is onto $(0, +\infty)$.

Remark 2. If circular orbits exist for (4), their motion always occurs with uniform angular speed. This follows from (7): If $r(t) = r_0$ is a constant solution of (8), then (7) implies that $\dot{\theta}(t)$ is constant.

Remark 3. Even when (8) admits constant solutions $r(t) = r_0$, they may or may not be Lyapunov stable. It is convenient to characterize Lyapunov stability for the corresponding steady state $(u_0, 0)$ of system (11)–(12), where $u_0 = 1/r_0$. Linearizing system (11)–(12) at $(u_0, 0)$ yields the following Jacobian matrix:

$$J_0 = \begin{pmatrix} 0 & 1 \\ -1 + f'(u_0) & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -G''(u_0) & 0 \end{pmatrix}.$$

If $G''(u_0) = 1 - f'(u_0)$ is negative, then J_0 has a positive and a negative eigenvalue, and $(u_0, 0)$ is a saddle which is unstable. On the other hand, if $G''(u_0) = 1 - f'(u_0)$ is positive, then J_0 has a pair of purely imaginary eigenvalues, and $(u_0, 0)$ is a center. This center is always a Lyapunov stable steady state for system (11)–(12). Indeed, first note that the (conserved) function $E(u, v)$ has a critical point at $(u_0, 0)$, since

$$\nabla E(u_0, 0) = \begin{pmatrix} G'(u_0) \\ 0 \end{pmatrix} = \begin{pmatrix} u_0 - f(u_0) \\ 0 \end{pmatrix} = \mathbf{0},$$

and that the Hessian of E at $(u_0, 0)$ is

$$H_0 = \begin{pmatrix} G''(u_0) & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - f'(u_0) & 0 \\ 0 & 1 \end{pmatrix},$$

which is a positive definite matrix. Hence, the function $E(u, v)$ has a strict local minimum $E_0 := G(u_0)$ at $(u_0, 0)$, and since $E' = 0$ along all solutions of system (11)–(12), Lyapunov stability of $(u_0, 0)$ follows from Lyapunov's Stability Theorem.

Instead of **(SCO)**, we shall work under the following, slightly stronger assumption in what follows.

(SCO') The admissible class of functions $h(r)$ is such that for every positive m and every nonzero l_0 , there exist some $u_0 > 0$ and some E_0 such that

1. $G(u_0) = E_0$,
2. $G'(u_0) = 0$, and
3. $G''(u_0) > 0$.

Note that **(SCO')** is slightly stronger than **(SCO)** because we have excluded the nongeneric possibility that $G''(u_0) = 0$, which may still give rise to $(u_0, 0)$ being a Lyapunov stable steady state for system (11)–(12). The previous analysis does not allow us to conclude this, and a higher-order analysis of the function $E(u, v)$ near $(u_0, 0)$ would be required to determine stability or instability.

Remark 4. To conclude, we claim that an important class of functions $h(r)$ which consists of certain power laws always satisfies **(SCO')**. Suppose that $h(r)$ is a power law, i.e.,

$$h(r) = kr^p \text{ for some } k > 0 \text{ and some real } p.$$

Then

$$f(u) = \frac{km}{l_0^2} u^{-(p+2)},$$

$$G(u) = \begin{cases} \frac{u^2}{2} + \frac{km}{l_0^2} \frac{u^{-(p+1)}}{p+1} & \text{if } p \neq -1, \\ \frac{u^2}{2} - \frac{km}{l_0^2} \ln u & \text{if } p = -1, \end{cases} \quad \text{and } G'(u) = u - \frac{km}{l_0^2} u^{-(p+2)}.$$

Note that the equation $G'(u) = 0$ has a positive solution u_0 for any value of m and l_0 if and only if $p \neq -3$; this solution u_0 is unique and given by

$$u_0 = \left(\frac{km}{l_0^2} \right)^{\frac{1}{p+3}} \text{ for } p \neq -3$$

and, furthermore,

$$G''(u_0) = p + 3,$$

which is positive if and only if $p > -3$. Thus, we have shown that the class of power laws

$$(13) \quad \mathcal{C}_{p>-3} = \{h(r) \mid h(r) = kr^p \text{ for some } k > 0 \text{ and some } p > -3\}$$

satisfies **(SCO')** and, furthermore, that any power law $h(r) = kr^p$ not in $\mathcal{C}_{p>-3}$ does not satisfy **(SCO')**. Note that $\mathcal{C}_{p>-3}$ contains Newtonian ($p = -2$) and Hookean ($p = 1$) gravitation. As a final remark, we note that if $h(r)$ belongs to $\mathcal{C}_{p>-3}$, then $G''(u_0)$ does not depend on u_0 . In other words, although the radius $r_0 = 1/u_0$ of the circular orbit $r(t) = r_0$ of (8) depends on, and varies with, the model parameter m and initial condition l_0 , the quantity $G''(u_0)$ does not depend on m or l_0 , nor on u_0 or on the value of the radius $r_0 = 1/u_0$.

3.2. Behavior of Solutions near the Stable Circular Orbit. If $h(r)$ is such that **(SCO')** holds, then we have seen that system (11)–(12) has a Lyapunov stable steady state $(u_0, 0)$, and that the function $E(u, v)$ is conserved along solutions of this system. Moreover, $E(u, v)$ attains a strict local minimum E_0 at $(u_0, 0)$ because the Hessian at $(u_0, 0)$ is positive definite. Consequently, for all $E > E_0$ sufficiently close to E_0 , the level curve $L_E = \{(u, v) \mid E(u, v) = E\}$ is a closed curve that coincides with the orbit of some solution $(u(\theta), v(\theta))$ of system (11)–(12) for which $E(u(\theta), v(\theta)) = E$ for all θ . Since L_E is a closed curve, the corresponding solution $(u(\theta), v(\theta))$ is a periodic solution of system (11)–(12). We can represent the minimal period of this solution as a convergent improper Riemann integral. In fact, because of the symmetry of the function $E(u, v)$ (namely, $E(u, v) = E(u, -v)$ for all (u, v)), and because the slopes to the solution through any pair of points (u, v) and $(u, -v)$ are opposite, we give a representation for half of the minimal period:

$$(14) \quad \Theta(E) := \int_{u_{\min}(E)}^{u_{\max}(E)} \frac{du}{\sqrt{2(E - G(u))}} \text{ for all } E > E_0 \text{ sufficiently close to } E_0.$$

Here, $u_{\min}(E)$ and $u_{\max}(E)$ are the minimal, respectively, maximal value of $u(\theta)$ for the solution $(u(\theta), v(\theta))$ whose orbit is the level curve L_E . These minimal and maximal values must be achieved at u -values where the orbit meets the u -axis (because the system equations (11)–(12) imply that if $u' = 0$, then $v = 0$), and we can find them by solving the equation

$$G(u) = E$$

for u in terms of E for all $E > E_0$ sufficiently close to E . It follows from the conditions on $G(u)$ in **(SCO')** that $G(u)$ has a strict local minimum E_0 at $u = u_0$, and the implicit function theorem then implies that the above equation has two smooth solutions $u_{\min}(E)$ and $u_{\max}(E)$ for all $E > E_0$ sufficiently close to E_0 , such that $0 < u_{\min}(E) < u_0 < u_{\max}(E)$.

The value of $\Theta(E)$ has important consequences for the nature of the corresponding solution $\mathbf{r}(t)$ for system (4). More precisely, suppose that $(u(\theta), v(\theta))$ is a solution of system (11)–(12) with orbit L_E . Since L_E is a closed curve, and recalling that $r = 1/u$, the orbit of the corresponding solution $\mathbf{r}(t)$ of (4) is certainly bounded because $u_{\min}(E)$ is positive. More importantly, we can characterize when this orbit is closed.

LEMMA 5. *Assume that $h(r)$ is such that **(SCO')** holds. Let $E > E_0$ be sufficiently close to E_0 such that $(u(\theta), v(\theta))$ is a periodic solution of system (11)–(12) with minimal period $2\Theta(E)$, and suppose that $\mathbf{r}(t)$ is the corresponding solution of*

system (4). Then the orbit of $\mathbf{r}(t)$ is closed if and only if $\Theta(E)$ is a rational multiple of π .

Proof. Suppose that $\Theta(E) = \frac{p}{q}\pi$ for some positive integers p and q . Define

$$T(E) = \int_0^{2q\Theta(E)} \frac{m}{l_0} \frac{d\theta}{u^2(\theta)}.$$

We claim that the corresponding solution $(r(t), \theta(t))$ for system (5)–(6) is such that $(r(t + T(E)), \theta(t + T(E))) = (r(t), \theta(t) + 2p\pi)$ for all t . This in turn implies that $\mathbf{r}(t)$ is $T(E)$ -periodic, and hence its orbit is closed. To prove the claim, note that by integrating (7), we find that

$$T(E) = \int_t^{t+T(E)} dt = \int_{\theta(t)}^{\theta(t+T(E))} \frac{m}{l_0} \frac{d\theta}{u^2(\theta)} \text{ for all } t.$$

Since $u(\theta)$ is $2\Theta(E)$ -periodic, it is also $2q\Theta(E)$ -periodic, and hence

$$T(E) = \int_{\theta(t)}^{\theta(t)+2q\Theta(E)} \frac{m}{l_0} \frac{d\theta}{u^2(\theta)} \text{ for all } t.$$

Combining both expressions yields that

$$\int_{\theta(t)+2q\Theta(E)}^{\theta(t+T(E))} \frac{m}{l_0} \frac{d\theta}{u^2(\theta)} = 0 \text{ for all } t,$$

and since the integrand of the integral above is positive for all θ , it follows that

$$\theta(t + T(E)) = \theta(t) + 2q\Theta(E) = \theta(t) + 2p\pi \text{ for all } t.$$

Moreover, this implies that

$$\begin{aligned} r(t + T(E)) &= \frac{1}{u(\theta(t + T(E)))} = \frac{1}{u(\theta(t) + 2p\pi)} = \frac{1}{u(\theta(t) + 2q\Theta(E))} = \frac{1}{u(\theta(t))} \\ &= r(t) \text{ for all } t, \end{aligned}$$

where we used the fact that $u(\theta)$ is $2\Theta(E)$ -periodic.

To prove the converse, suppose that $\mathbf{r}(t)$ is T -periodic. Then the corresponding solution $(r(t), \theta(t))$ of system (5)–(6) is T -periodic. More specifically, there is some integer k such that $(r(t + T), \theta(t + T)) = (r(t), \theta(t) + 2k\pi)$ for all t . Hence,

$$T = \int_t^{t+T} dt = \int_{\theta(t)}^{\theta(t+T)} \frac{m}{l_0} \frac{d\theta}{u^2(\theta)} = \int_{\theta(t)}^{\theta(t)+2k\pi} \frac{m}{l_0} \frac{d\theta}{u^2(\theta)} \text{ for all } t.$$

Differentiating with respect to t shows that $u(\theta(t) + 2k\pi) = u(\theta(t))$ for all t , and since the map $t \rightarrow \theta(t)$ is monotone and maps $\mathbb{R}$ onto $\mathbb{R}$, this implies that $u(\theta)$ is $2k\pi$ -periodic. Since the minimal period of $u(\theta)$ is $2\Theta(E)$, $2k\pi$ must be an integer multiple of $2\Theta(E)$. $\square$

Recall the representation $\Theta(E)$ in (14) for half of the minimal period of the periodic solution $(u(\theta), v(\theta))$ of system (11)–(12) whose orbit is L_E . From the theory of ordinary differential equations, it follows that the function $E \rightarrow \Theta(E)$ is continuous. We next seek to evaluate the limit of $\Theta(E)$ as $E \rightarrow E_0+$, and we will find that this limit only depends on the second derivative of the function G at u_0 .

LEMMA 6. Assume that $h(r)$ is such that **(SCO')** holds. For all $E > E_0$ sufficiently close to E_0 , let $(u(\theta), v(\theta))$ be the periodic solution of system (11)–(12) with orbit L_E and minimal period $2\Theta(E)$. Then

$$\Theta_0 := \lim_{E \rightarrow E_0+} \Theta(E) = \lim_{E \rightarrow E_0+} \int_{u_{\min}(E)}^{u_{\max}(E)} \frac{du}{\sqrt{2(E - G(u))}}$$

exists and

$$(15) \quad \Theta_0 = \frac{\pi}{\sqrt{G''(u_0)}}.$$

Proof. By a translation of the point (u_0, E_0) in the (u, E) -plane to the origin, it suffices to show that

$$(16) \quad \lim_{E \rightarrow 0+} \int_{u_-(E)}^{u_+(E)} \frac{du}{\sqrt{2(E - G(u))}} = \frac{\pi}{\sqrt{G''(0)}},$$

where G is a function with $G(0) = G'(0) = 0$ and $G''(0) > 0$, and $u_-(E)$ and $u_+(E)$ are the two solutions to the equation

$$G(u) = E,$$

defined for all $E > 0$ sufficiently close to $E = 0$ such that $u_-(E) < 0 < u_+(E)$ for all E , and $\lim_{E \rightarrow 0+} u_{\pm}(E) = 0$.

Step 1: Let us start by proving (16) for the special case that $G(u)$ is a quadratic function:

$$G(u) = \frac{G''(0)}{2} u^2,$$

with $G''(0) > 0$. First note that in this case, $u_-(E)$ and $u_+(E)$ can be calculated explicitly,

$$u_-(E) = -\sqrt{2E/G''(0)} \quad \text{and} \quad u_+(E) = +\sqrt{2E/G''(0)} \quad \text{for all } E > 0,$$

and thus for all $E > 0$,

$$\begin{aligned} \int_{u_-(E)}^{u_+(E)} \frac{du}{\sqrt{2(E - G(u))}} &= \int_{-\sqrt{2E/G''(0)}}^{\sqrt{2E/G''(0)}} \frac{du}{\sqrt{2E - G''(0)u^2}} \\ &= \frac{1}{\sqrt{G''(0)}} \int_{-1}^{+1} \frac{dv}{\sqrt{1 - v^2}} \\ &= \frac{\pi}{\sqrt{G''(0)}}. \end{aligned}$$

In particular, since this integral is independent of E , (16) follows in this special case.

Step 2: Given the above result in Step 1, in order to prove (16), it suffices to show that

$$(17) \quad \lim_{E \rightarrow 0+} \int_0^{u_+(E)} \frac{du}{\sqrt{2(E - G(u))}} - \int_0^{\sqrt{2E/G''(0)}} \frac{du}{\sqrt{2(E - G''(0)u^2/2)}} = 0,$$

because a similar result can be obtained for the difference of the integral between $u_-(E)$ and 0, and the integral between $-\sqrt{2E/G''(0)}$ and 0. By two substitutions, (17) is equivalent to

$$(18) \quad \lim_{E \rightarrow 0+} \int_0^1 \frac{u_+(E)}{\sqrt{2E}} \frac{1}{\sqrt{1 - G(u_+(E)x)/E}} - \frac{1}{\sqrt{G''(0)}} \frac{1}{\sqrt{1 - x^2}} dx = 0.$$

Denote the integrand in (18) by $f(x, E)$. We shall prove (18) by applying the dominated convergence theorem. To that end, we must show that

1. $\lim_{E \rightarrow 0+} f(x, E) = 0$ for all x in $(0, 1)$, and
2. there exists a function $g(x)$ defined for x in $(0, 1)$ such that $\int_0^1 g(x) dx$ is finite, and

$$|f(x, E)| \leq g(x) \text{ for all } x \text{ in } (0, 1) \text{ and for all sufficiently small } E > 0.$$

To prove the first result, observe that we can choose $\bar{E} > 0$ sufficiently small so that $G(u) > 0$, $G'(u) > 0$, and $G''(u) \geq \alpha > 0$ for all u in $(0, u_+(E))$ and all E in $(0, \bar{E})$ (the existence of such a positive α is guaranteed because G is C^2 and $G''(0) > 0$). As $G(u_+(E)) = E$, it follows from Taylor's Theorem that

$$E = G(u_+(E)) = G''(\zeta(E)) \frac{u_+^2(E)}{2} \text{ for all } E \text{ in } (0, \bar{E}), \text{ where } 0 < \zeta(E) < u_+(E).$$

Similarly, for all (x, E) in $(0, 1) \times (0, \bar{E})$, Taylor's Theorem implies that

$$G(u_+(E)x) = G''(\xi(x, E)) \frac{(u_+(E)x)^2}{2} \text{ for some } 0 < \xi(x, E) < u_+(E)x.$$

These two equations imply that for all (x, E) in $(0, 1) \times (0, \bar{E})$,

$$\begin{aligned} \frac{u_+(E)}{\sqrt{2E}} \frac{1}{\sqrt{1 - G(u_+(E)x)/E}} &= \frac{1}{\sqrt{G''(\zeta(E))}} \frac{1}{\sqrt{1 - \frac{G''(\xi(x, E))}{G''(\zeta(E))} x^2}} \\ &\rightarrow \frac{1}{\sqrt{G''(0)}} \frac{1}{\sqrt{1 - x^2}} \text{ as } E \rightarrow 0+, \end{aligned}$$

because G is C^2 , and $\zeta(E) \rightarrow 0$ and $\xi(x, E) \rightarrow 0$ as $E \rightarrow 0+$ for all x in $(0, 1)$. This establishes the first result.

To prove the second result, first note that for all (x, E) in $(0, 1) \times (0, \bar{E})$,

$$|f(x, E)| \leq \frac{u_+(E)}{\sqrt{2E}} \frac{1}{\sqrt{1 - G(u_+(E)x)/E}} + \frac{1}{\sqrt{G''(0)}} \frac{1}{\sqrt{1 - x^2}},$$

and that the second term on the right is a function of x only and is integrable over the interval $[0, 1]$. Therefore, we focus on the first term and claim that there exists $\beta > 0$ such that for all (x, E) in $(0, 1) \times (0, \bar{E})$,

$$(19) \quad \frac{u_+(E)}{\sqrt{2E}} \frac{1}{\sqrt{1 - G(u_+(E)x)/E}} \leq \frac{\beta}{\sqrt{x(1-x)}}.$$

From this, the second result follows because the function on the right is integrable over the interval $[0, 1]$. Indeed, by the substitution $v = \sqrt{x}$, we see that

$$\int_0^1 \frac{\beta}{\sqrt{x(1-x)}} dx = 2 \int_0^1 \frac{\beta}{\sqrt{1-v^2}} dv = \beta\pi.$$

To prove (19), first note that for all (x, E) in $(0, 1) \times (0, \bar{E})$,

$$(20) \quad \frac{u_+(E)}{\sqrt{2E}} \frac{1}{\sqrt{1 - G(u_+(E)x)/E}} = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{\frac{G(u_+(E)) - G(u_+(E)x)}{u_+^2(E)}}}.$$

Since $G(u)$ is strictly convex for u in $(0, u_+(E))$ and all E in $(0, \bar{E})$, we have that

$$G(u_+(E)) > G(u_+(E)x) + G'(u_+(E)x)(u_+(E) - u_+(E)x)$$

for all (x, E) in $(0, 1) \times (0, \bar{E})$. From this, we obtain that

$$0 < \frac{G'(u_+(E)x)(1-x)}{u_+(E)} < \frac{G(u_+(E)) - G(u_+(E)x)}{u_+^2(E)} \text{ for all } (x, E) \text{ in } (0, 1) \times (0, \bar{E}),$$

and inserting this in (20), we obtain that for all (x, E) in $(0, 1) \times (0, \bar{E})$,

$$\begin{aligned} \frac{u_+(E)}{\sqrt{2E}} \frac{1}{\sqrt{1 - G(u_+(E)x)/E}} &< \frac{1}{\sqrt{2}} \sqrt{\frac{u_+(E)}{G'(u_+(E)x)}} \frac{1}{\sqrt{1-x}} \\ &= \frac{1}{\sqrt{2}} \frac{1}{\sqrt{G''(\nu(x, E))}} \frac{1}{\sqrt{x(1-x)}}, \end{aligned}$$

where $0 < \nu(x, E) < u_+(E)x$ by Taylor's Theorem applied to $G'(u)$. Recalling that $G'(u) \geq \alpha > 0$ for all u in $(0, u_+(E))$ and all E in $(0, \bar{E})$, our claim is proved when we set

$$\beta = \frac{1}{\sqrt{2\alpha}}. \quad \square$$

The next result addresses the question of when Θ_0 is independent of u_0 or, equivalently, when Θ_0 is independent of the radius of the stable circular orbit $r_0 = 1/u_0$. It states that this happens only if $h(r)$ belongs to a specific class of power laws.

LEMMA 7. *Assume that the conditions of Lemma 6 hold. Then Θ_0 is independent of u_0 only if $h(r)$ belongs to $\mathcal{C}_{p>-3}$, the class of power laws that was defined in (13).*

Proof. Let $u_0 > 0$ be such that $G'(u_0) = 0$. Then $G''(u_0) = 1 - f'(u_0)$ is independent of u_0 only if $1 - f'(u_0) = c$ for some positive constant c . As $G'(u_0) = u_0 - f(u_0) = 0$, $f(u_0)$ must satisfy

$$1 - u_0 \frac{f'(u_0)}{f(u_0)} = c.$$

Solving this equation for $f(u)$ yields that

$$f(u) = \kappa u^{1-c} \text{ for some } \kappa > 0,$$

and recalling that $u = 1/r$ and that $f(u)$ and $h(r)$ are related by (10), it follows that

$$h(r) = \frac{l_0}{m} \kappa r^{c-3}.$$

But $c > 0$, and hence $h(r)$ must belong to $\mathcal{C}_{p>-3}$, defined in (13). $\square$

3.3. Orbits with Large Energy When $h(r)$ Belongs to $\mathcal{C}_{p>-3}$. Working under the assumption (SCO'), our focus up to this point has been on the behavior of solutions of system (11)–(12) near the steady state $(u_0, 0)$ or, equivalently, on the behavior of solutions of (8) near the circular orbit corresponding to the solution $r(r) = r_0$, where $r_0 = 1/u_0$. Here we shift attention in order to understand the behavior of bounded solutions of system (11)–(12) corresponding to large values E of the conserved energy function $E(u, v)$ in the case that the gravitational law $h(r)$ belongs to $\mathcal{C}_{p>-3}$. We first collect some basic results to prepare ourselves for this problem.

LEMMA 8. Assume that $h(r)$ belongs to $\mathcal{C}_{p>-3}$. Then $G : (0, +\infty) \rightarrow \mathbb{R}$, defined by

$$G(u) = \begin{cases} \frac{u^2}{2} + \frac{km}{l_0^2} \frac{u^{-(p+1)}}{p+1} & \text{if } p \neq 1 \text{ (and } p > -3), \\ \frac{u^2}{2} - \frac{km}{l_0^2} \ln u & \text{if } p = -1, \end{cases}$$

has a unique global minimum $E_0 = G(u_0)$ at $u = u_0$, where $u_0 = (km/l_0^2)^{\frac{1}{p+3}}$, and $G(u)$ is decreasing for u in $(0, u_0)$ and increasing for $u > u_0$. Furthermore,

- if $p \in (-3, -1)$, then $\lim_{u \rightarrow 0+} G(u) = 0 = G(\bar{u})$, where $\bar{u} = \left(\frac{2km}{-(p+1)l_0^2}\right)^{\frac{1}{p+3}} > 0$;
- if $p \geq -1$, then $\lim_{u \rightarrow 0+} G(u) = +\infty = \lim_{u \rightarrow +\infty} G(u)$.

The two solutions $u_{\min}(E)$ and $u_{\max}(E)$ of the equation $G(u) = E$ are defined

- for all $E \in (E_0, 0)$ if $p \in (-3, -1)$; $u_{\min}(E)$ decreases and $u_{\min}(E) \rightarrow 0$ as $E \rightarrow 0^-$; $u_{\max}(E)$ increases and $u_{\max}(E) \rightarrow \bar{u}$ as $E \rightarrow 0^-$;
- for all $E \in (E_0, +\infty)$ if $p \geq -1$; $u_{\min}(E)$ decreases and $u_{\min}(E) \rightarrow 0$ as $E \rightarrow +\infty$; $u_{\max}(E)$ increases and $u_{\max}(E) \rightarrow +\infty$ as $E \rightarrow +\infty$.

Let $y_1(E) = u_{\min}(E)/u_{\max}(E)$, and define

$$(21) \quad H(y, E) = \begin{cases} \frac{2E}{u_{\max}^2(E)}(1 - y^{-(p+1)}) + y^{-(p+1)} - y^2 & \text{for } p \neq -1, \\ \left(1 - \frac{2E}{u_{\max}^2(E)}\right) \frac{\ln y}{\ln(u_{\max}(E))} + 1 - y^2 & \text{for } p = -1 \end{cases}$$

for all (y, E) in $(y_1(E), 1) \times (E_0, 0)$ if $p \in (-3, -1)$, and for all (y, E) in $(y_1(E), 1) \times (E_0, +\infty)$ if $p \geq -1$. Then half of the minimal period of the periodic solution $(u(\theta), v(\theta))$ of system (11)–(12) with orbit L_E is given by

$$(22) \quad \Theta(E) = \int_{y_1(E)}^1 \frac{dy}{\sqrt{H(y, E)}}.$$

Proof. For $h(r)$ belonging to $\mathcal{C}_{p>-3}$, the function $G(u)$ was introduced in Remark 4 of section 3.1. The statements regarding the monotonicity of G , and its limiting behavior when $u \rightarrow 0+$ and $u \rightarrow +\infty$, are routine exercises in calculus which are left to the reader. The existence of the globally defined functions $u_{\min}(E)$ and $u_{\max}(E)$ on the appropriate domains for E (which depend on the value of p) are implied by the above properties of $G(u)$. Formula (14) for half of the minimal period of the periodic solution $(u(\theta), v(\theta))$ with orbit L_E then extends to these domains of E , and the substitution $y = u/u_{\max}(E)$ in (14) together with the fact that $G(u_{\max}(E)) = E$ lead to the representation (22). $\square$

Remark. Suppose that $h(r)$ belongs to $\mathcal{C}_{p>-3}$. If $p \geq -1$, then for every $E \geq E_0$ the solution $(u(\theta), v(\theta))$ of system (11)–(12) with orbit L_E satisfies that $u(\theta) \geq u_{\min}(E) > 0$ for all θ . Consequently, all solutions $\mathbf{r}(t)$ of (4) are bounded (recall that $r = 1/u$, hence $r(t) \leq 1/u_{\min}(E)$ for all t). The same is true when $-3 < p < -1$,

provided that the solution $\mathbf{r}(t)$ corresponds to a solution of (11)–(12) with an orbit L_E for which $E \in (E_0, 0)$. But what happens when $E \geq 0$? We leave it as an exercise to the reader to verify that all such solutions $\mathbf{r}(t)$ are unbounded. This follows from a phase plane analysis of system (11)–(12) or, equivalently, by examining the level curves L_E of the function $E(u, v)$ for values $E \geq 0$ and noting that the closure of these curves always contains points of the form $(0, v)$ for some real v .

Under the assumptions of Lemma 8, our main objective is to compute the following limits:

$$\lim_{E \rightarrow 0^-} \Theta(E) \text{ for } p \in (-3, -1) \text{ and } \lim_{E \rightarrow +\infty} \Theta(E) \text{ for } p \geq -1.$$

We shall start with the latter limit. Its value was stated in [1] without proof, and a proof was given only recently in [4] by Jovanović. To make this paper self-contained, we include Jovanović's proof.

LEMMA 9. *Assume that the conditions and notation of Lemma 8 hold. If $p \geq -1$, then*

$$(23) \quad \lim_{E \rightarrow +\infty} \Theta(E) = \lim_{E \rightarrow +\infty} \int_{y_1(E)}^1 \frac{dy}{\sqrt{H(y, E)}} = \frac{\pi}{2}.$$

Proof. We only prove the case that $p > -1$, because the proof for the case $p = -1$ is similar.

Let us start by examining the properties of the function $H(y, E)$. For each E in (E_0, ∞) , $H(y, E)$ is positive for $y \in (y_1(E), 1)$ and zero at the endpoints $y = y_1(E)$ and $y = 1$. Furthermore, $H(y, E)$ has a unique maximum at $y = \bar{y}(E)$, where

$$(24) \quad \bar{y}(E) = \left(\frac{p+1}{2} \left[\frac{2E}{u_{\max}^2(E)} - 1 \right] \right)^{\frac{1}{p+3}} = \left(\frac{km}{l_0^2} \right)^{\frac{1}{p+3}} u_{\max}^{-1}(E),$$

and where we used that fact that $G(u_{\max}(E)) = E$ for all E implies that

$$(25) \quad \left[\frac{2E}{u_{\max}^2(E)} - 1 \right] = \frac{2km}{l_0^2(p+1)} u_{\max}^{-(p+3)}(E).$$

For each E in $(E_0, +\infty)$, $H(y, E)$ is strictly increasing in y on $[y_1(E), \bar{y}(E)]$ and strictly decreasing in y on $[\bar{y}(E), 1]$, and it is strictly concave for y in $[y_1(E), 1]$ because

$$\begin{aligned} \frac{\partial^2 H}{\partial y^2} &= -(p+1)(p+2) \left[\frac{2E}{u_{\max}^2(E)} - 1 \right] y^{-(p+3)} - 2 \\ &= -2 \left(\frac{km}{l_0^2} (p+2) u_{\max}^{-(p+3)}(E) y^{-(p+3)} + 1 \right) \end{aligned}$$

is negative (again we used (25)). These properties of $H(y, E)$ are illustrated in Figure 1. Note for future reference that $y_1(E) \rightarrow 0$ and $\bar{y}(E) \rightarrow 0$ as $E \rightarrow +\infty$ because $u_{\min}(E) \rightarrow 0$ and $u_{\max}(E) \rightarrow +\infty$ as $E \rightarrow +\infty$ according to Lemma 8. Finally, using the definition of $H(y, E)$, (24), and (25), it follows that $H(\bar{y}(E), E) \rightarrow 1$ as $E \rightarrow +\infty$.

Recall from calculus that

$$\int_0^1 \frac{dy}{\sqrt{1-y^2}} = \frac{\pi}{2},$$

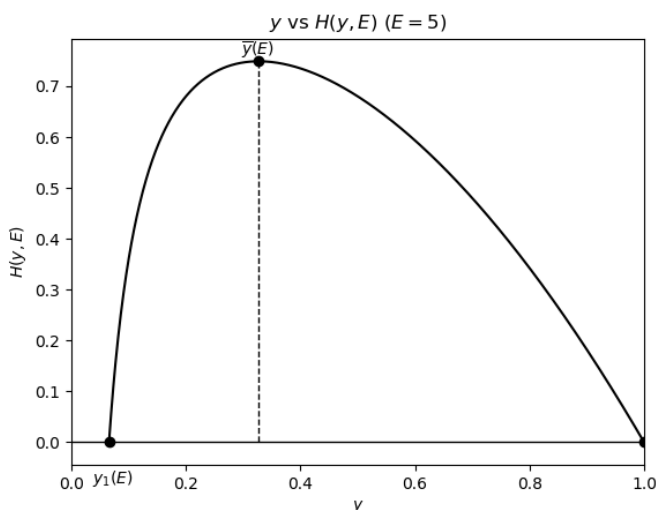


Fig. 1 Graph of $H(y, E)$ when $p \geq -1$.

and therefore it holds for all $E > E_0$ that

$$\left| \Theta(E) - \frac{\pi}{2} \right| \leq \left| \int_{y_1(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{1-y^2}} dy \right| + \int_0^{y_1(E)} \frac{dy}{\sqrt{1-y^2}}.$$

The second term on the right-hand side converges to 0 as $E \rightarrow +\infty$. Thus, to prove (23), it suffices to prove that the first term on the right-hand side converges to zero. This happens provided that the following three limits can be established:

$$(26) \quad \lim_{E \rightarrow +\infty} \int_{y_1(E)}^{\bar{y}(E)} \frac{dy}{\sqrt{1-y^2}} = 0,$$

$$(27) \quad \lim_{E \rightarrow +\infty} \int_{y_1(E)}^{\bar{y}(E)} \frac{dy}{\sqrt{H(y, E)}} = 0,$$

$$(28) \quad \lim_{E \rightarrow +\infty} \int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{1-y^2}} dy = 0.$$

Note that (26) holds because $y_1(E) \rightarrow 0$ and $\bar{y}(E) \rightarrow 0$ as $E \rightarrow +\infty$.

To show (27), note that the strict concavity of $H(y, E)$ in terms of y for each E implies that

$$H(y, E) \geq \frac{H(\bar{y}(E), E)}{\bar{y}(E) - y_1(E)}(y - y_1(E)) \text{ for all } y \in [y_1(E), \bar{y}(E)]$$

and thus

$$\begin{aligned} \int_{y_1(E)}^{\bar{y}(E)} \frac{dy}{\sqrt{H(y, E)}} &\leq \sqrt{\frac{\bar{y}(E) - y_1(E)}{H(\bar{y}(E), E)}} \int_{y_1(E)}^{\bar{y}(E)} \sqrt{y - y_1(E)} dy \\ &= \sqrt{\frac{\bar{y}(E) - y_1(E)}{H(\bar{y}(E), E)}} \frac{(\bar{y}(E) - y_1(E))^{3/2}}{3/2} \rightarrow 0 \text{ as } E \rightarrow +\infty. \end{aligned}$$

It remains to prove (28). Note that using (25),

$$\begin{aligned} \int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{1-y^2}} dy &= \int_{\bar{y}(E)}^1 \frac{1-y^2-H(y, E)}{\sqrt{H(y, E)}\sqrt{1-y^2}(\sqrt{H(y, E)}+\sqrt{1-y^2})} dy \\ &= \left[\frac{2E}{u_{\max}^2(E)} - 1 \right] \\ &\quad \times \int_{\bar{y}(E)}^1 \frac{y^{-(p+1)}-1}{\sqrt{H(y, E)}\sqrt{1-y^2}(\sqrt{H(y, E)}+\sqrt{1-y^2})} dy \\ &= \frac{2km}{l_0^2} \frac{u_{\max}^{-(p+3)}(E)}{p+1} \\ &\quad \times \int_{\bar{y}(E)}^1 \frac{y^{-(p+1)}-1}{\sqrt{H(y, E)}\sqrt{1-y^2}(\sqrt{H(y, E)}+\sqrt{1-y^2})} dy \\ &\leq \frac{2km}{l_0^2} \frac{u_{\max}^{-(p+3)}(E)}{p+1} \int_{\bar{y}(E)}^1 \frac{y^{-(p+1)}-1}{(H(y, E))^{3/2}} dy, \end{aligned}$$

because $H(y, E) \leq 1-y^2$ when $y \in [\bar{y}(E), 1]$.

Strict concavity of $H(y, E)$ in terms of y for each E implies that

$$H(y, E) \geq \frac{H(\bar{y}(E), E)}{1-\bar{y}(E)}(1-y) \text{ for all } y \in [\bar{y}(E), 1],$$

and consequently

$$\begin{aligned} &\int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{1-y^2}} dy \\ &\leq \frac{2km}{l_0^2} \frac{u_{\max}^{-(p+3)}(E)}{p+1} \left(\frac{1-\bar{y}(E)}{H(\bar{y}(E), E)} \right)^{3/2} \int_{\bar{y}(E)}^1 \frac{y^{-(p+1)}-1}{1-y} \frac{dy}{\sqrt{1-y}}. \end{aligned}$$

Since $\bar{y}(E) \rightarrow 0$ as $E \rightarrow +\infty$, it follows that $\bar{y}(E) < 1/2$ for all sufficiently large E , and hence we can break up the integral on the right-hand side into two parts when E is sufficiently large:

$$\begin{aligned} \int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{1-y^2}} dy &\leq \frac{2km}{l_0^2(p+1)} \left(\frac{1-\bar{y}(E)}{H(\bar{y}(E), E)} \right)^{3/2} \\ &\quad \times \left(u_{\max}^{-(p+3)}(E) \int_{\bar{y}(E)}^{1/2} \frac{y^{-(p+1)}-1}{1-y} \frac{dy}{\sqrt{1-y}} \right. \\ &\quad \left. + u_{\max}^{-(p+3)}(E) \int_{1/2}^1 \frac{y^{-(p+1)}-1}{1-y} \frac{dy}{\sqrt{1-y}} \right) \\ &\leq \frac{2km}{l_0^2(p+1)} \left(\frac{1-\bar{y}(E)}{H(\bar{y}(E), E)} \right)^{3/2} \\ &\quad \times \left(u_{\max}^{-(p+3)}(E) \int_{\bar{y}(E)}^{1/2} \frac{\bar{y}^{-(p+1)}(E)}{(1-y)^{3/2}} dy \right. \\ &\quad \left. + u_{\max}^{-(p+3)}(E) \int_{1/2}^1 \frac{y^{-(p+1)}-1}{1-y} \frac{dy}{\sqrt{1-y}} \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{2km}{l_0^2(p+1)} \left(\frac{1 - \bar{y}(E)}{H(\bar{y}(E), E)} \right)^{3/2} \\
&\quad \times \left((km/l_0^2)^{-(p+1)/(p+3)} u_{\max}^{-2}(E) \int_{\bar{y}(E)}^{1/2} \frac{dy}{(1-y)^{3/2}} \right. \\
&\quad \left. + u_{\max}^{-(p+3)}(E) \int_{1/2}^1 \frac{y^{-(p+1)} - 1}{1-y} \frac{dy}{\sqrt{1-y}} \right).
\end{aligned}$$

Here we used the fact that the map $y \rightarrow y^{-(p+1)} - 1$ is decreasing on $[\bar{y}(E), 1/2]$ and thus certainly bounded by $\bar{y}^{-(p+1)}(E)$ to obtain the second inequality, and we also used (24) to obtain the last equality. Note that

$$\int_{\bar{y}(E)}^{1/2} \frac{dy}{(1-y)^{3/2}} \leq \int_0^{1/2} \frac{dy}{(1-y)^{3/2}} = 2(\sqrt{2} - 1),$$

and, furthermore, that the function $y \rightarrow (y^{-(p+1)} - 1)/(1-y)$ is bounded on the interval $[1/2, 1]$ (L'Hôpital's rule shows that it has limit equal to $p+1$ as $y \rightarrow 1$). Letting $M > 0$ be a bound for this function, it follows that

$$\int_{1/2}^1 \frac{y^{-(p+1)} - 1}{1-y} \frac{dy}{\sqrt{1-y}} \leq M \int_{1/2}^1 \frac{dy}{\sqrt{1-y}} = \sqrt{2}M.$$

These two facts, and using that $\bar{y}(E) \rightarrow 0$, $H(\bar{y}(E), E) \rightarrow 1$, $u_{\max}(E) \rightarrow +\infty$ as $E \rightarrow +\infty$, and that $p > -1$, imply that (28) holds. $\square$

We now turn to the calculation of the limit

$$\lim_{E \rightarrow 0-} \Theta(E) \text{ for } p \in (-3, -1).$$

The value of this limit was given in [1] without proof. It turns out that the methods developed by Jovanović in the proof of Lemma 9 can be adapted to establish the value of the limit when $p \in [-2, -1)$, but new arguments are needed in the case $p \in (-3, -2)$. That is why we split the proof up into two cases, and we shall start with the former case. Before we start, we recall the following integral from calculus, which is easily computed using the substitution $x = y^{(p+3)/2}$:

$$(29) \quad \int_0^1 \frac{dy}{\sqrt{y^{-(p+1)} - y^2}} = \frac{\pi}{p+3} \text{ for all } p > -3.$$

LEMMA 10. Assume that the conditions and notation of Lemma 8 hold. If $p \in [-2, -1)$, then

$$(30) \quad \lim_{E \rightarrow 0-} \Theta(E) = \lim_{E \rightarrow 0-} \int_{y_1(E)}^1 \frac{dy}{\sqrt{H(y, E)}} = \frac{\pi}{p+3}.$$

Proof. We start by examining the properties of the function $H(y, E)$. For each E in $(E_0, 0)$, $H(y, E)$ is positive for $y \in (y_1(E), 1)$ and zero at the endpoints $y = y_1(E)$ and $y = 1$. Furthermore, $H(y, E)$ has a unique maximum at $y = \bar{y}(E)$, where

$$(31) \quad \bar{y}(E) = \left(\frac{p+1}{2} \left[\frac{2E}{u_{\max}^2(E)} - 1 \right] \right)^{\frac{1}{p+3}} = \left(\frac{km}{l_0^2} \right)^{\frac{1}{p+3}} u_{\max}^{-1}(E),$$

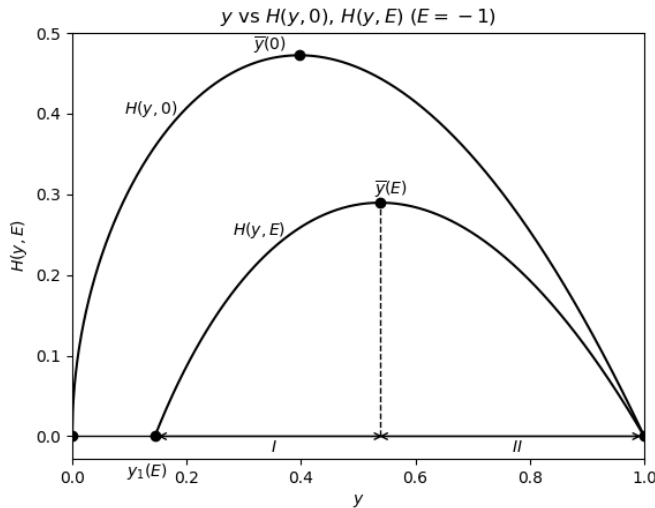


Fig. 2 Graphs of $H(y, E)$ and $H(y, 0)$ when $-2 \leq p < -1$.

and where we used that fact that $G(u_{\max}(E)) = E$ for all E , which implies that

$$(32) \quad \left[\frac{2E}{u_{\max}^2(E)} - 1 \right] = \frac{2km}{l_0^2(p+1)} u_{\max}^{-(p+3)}(E).$$

For each E in $(E_0, 0)$, $H(y, E)$ is strictly increasing in y on $[y_1(E), \bar{y}(E)]$ and strictly decreasing in y on $[\bar{y}(E), 1]$, and it is strictly concave for y in $[y_1(E), 1]$ because

$$\begin{aligned} \frac{\partial^2 H}{\partial y^2} &= -(p+1)(p+2) \left[\frac{2E}{u_{\max}^2(E)} - 1 \right] y^{-(p+3)} - 2 \\ &= -2 \left(\frac{km}{l_0^2} (p+2) u_{\max}^{-(p+3)}(E) y^{-(p+3)} + 1 \right) \end{aligned}$$

is negative (again we used (32)). These properties of $H(y, E)$ are illustrated in Figure 2.

Note for future reference that $y_1(E) \rightarrow 0$ and $\bar{y}(E) \rightarrow (km/l_0^2)^{1/(p+3)} \bar{u}^{-1} := \bar{y}(0) > 0$ as $E \rightarrow 0^-$ because $u_{\min}(E) \rightarrow 0$ and $u_{\max}(E) \rightarrow \bar{u}$ as $E \rightarrow 0^-$ according to Lemma 8. For ease of notation, we define $H(y, 0) := y^{-(p+1)} - y^2$ for all $y \in [0, 1]$, whose graph is also illustrated in Figure 2. Then, using the definition of $H(y, E)$, (31), and (32), it follows that $H(\bar{y}(E), E) \rightarrow H(\bar{y}(0), 0) > 0$ as $E \rightarrow +\infty$.

By (29) and the definition of $H(y, 0)$, we see that for all $E \in (E_0, 0)$,

$$\left| \Theta(E) - \frac{\pi}{p+3} \right| \leq \left| \int_{y_1(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy \right| + \int_0^{y_1(E)} \frac{dy}{\sqrt{H(0, y)}}.$$

The second term on the right-hand side can be evaluated using the substitution $x = y^{(p+3)/2}$,

$$\int_0^{y_1(E)} \frac{dy}{\sqrt{H(0, y)}} = \frac{2}{p+3} \arcsin \left(y_1^{(p+3)/2}(E) \right) \rightarrow 0 \text{ as } E \rightarrow 0^-,$$

because $y_1(E) \rightarrow 0$ as $E \rightarrow 0-$. Thus, to prove (23), it suffices to prove that the first term on the right-hand side converges to zero, which will be the case if we can establish the following two limits:

$$(33) \quad \lim_{E \rightarrow 0-} \int_{y_1(E)}^{\bar{y}(E)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy = 0,$$

$$(34) \quad \lim_{E \rightarrow 0-} \int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy = 0.$$

Let us begin with (33). By strict concavity of both $H(y, E)$ and $H(y, 0)$, we have that for all E in $(E_0, 0)$ and for all y in $[y_1(E), \bar{y}(E)]$,

$$\begin{aligned} H(y, E) &\geq \frac{H(\bar{y}(E), E)}{\bar{y}(E) - y_1(E)}(y - y_1(E)) \text{ and} \\ H(y, 0) &\geq \frac{H(\bar{y}(E), 0) - H(y_1(E), 0)}{\bar{y}(E) - y_1(E)}(y - y_1(E)) + H(y_1(E), 0). \end{aligned}$$

Then

$$\begin{aligned} \int_{y_1(E)}^{\bar{y}(E)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy &\leq -\frac{2E}{u_{\max}^2(E)} \int_{y_1(E)}^{\bar{y}(E)} \frac{1 - y^{-(p+1)}}{\sqrt{H(y, E)H(y, 0)}} dy \\ &\leq -\frac{2E}{u_{\max}^2(E)} \frac{\alpha(E)}{\sqrt{\frac{H(\bar{y}(E), E)}{\bar{y}(E) - y_1(E)} H(y_1(E), 0)}} \\ &\quad \times \int_{y_1(E)}^{\bar{y}(E)} \frac{dy}{\sqrt{y - y_1(E)}(y - y_1(E) + \alpha(E))}, \end{aligned}$$

where we define $\alpha(E) = \frac{H(y_1(E), 0)(\bar{y}(E) - y_1(E))}{H(\bar{y}(E), 0) - H(y_1(E), 0)}$ for all E in $(E_0, 0)$. Note that $\alpha(E)$ is positive and continuous, and that $\lim_{E \rightarrow 0-} \alpha(E) = 0$. Making the substitution $x = \sqrt{(y - y_1(E))/\alpha(E)}$ in the last integral yields

$$\begin{aligned} \int_{y_1(E)}^{\bar{y}(E)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy &\leq -\frac{2E}{u_{\max}^2(E)} \frac{\alpha(E)}{\sqrt{\frac{H(\bar{y}(E), E)}{\bar{y}(E) - y_1(E)} H(y_1(E), 0)}} \\ &\quad \times \frac{2}{\sqrt{\alpha(E)}} \arctan \left(\sqrt{\frac{\bar{y}(E) - y_1(E)}{\alpha(E)}} \right) \\ &\leq -\frac{2E}{u_{\max}^2(E)} \frac{\sqrt{\alpha(E)}}{\sqrt{\frac{H(\bar{y}(E), E)}{\bar{y}(E) - y_1(E)} H(y_1(E), 0)}} \pi \\ &= \frac{-E}{\sqrt{H(y_1(E), 0)}} \\ &\quad \times \frac{2\pi(\bar{y}(E) - y_1(E))}{u_{\max}^2(E) \sqrt{H(\bar{y}(E), E) (H(\bar{y}(E), 0) - H(y_1(E), 0))}}. \end{aligned}$$

Note that the second factor in the last expression converges to a finite value as $E \rightarrow 0-$ because $u_{\max}(E) \rightarrow \bar{u} > 0$, $\bar{y}(E) \rightarrow \bar{y}(0) > 0$, $y_1(E) \rightarrow 0$, $H(\bar{y}(E), E) \rightarrow H(\bar{y}(0), 0) > 0$, $H(\bar{y}(E), 0) \rightarrow H(\bar{y}(0), 0) > 0$, and $H(y_1(E), 0) \rightarrow 0$. Moreover, the first factor

converges to zero:

$$\begin{aligned} \frac{-E}{\sqrt{H(y_1(E), 0)}} &= \frac{-E}{y_1^{-\frac{p+1}{2}}(E)} \frac{1}{\sqrt{1 - y_1^{p+3}(E)}} \\ &= -y_1^{-\frac{p+1}{2}}(E) \frac{\frac{1}{2} u_{\max}^2(E) y_1^{p+3}(E) + \frac{km}{l_0^2(p+1)} u_{\max}^{-(p+1)}(E)}{\sqrt{1 - y_1^{p+3}(E)}} \\ &\rightarrow 0 \text{ as } E \rightarrow 0-, \end{aligned}$$

where we used that $G(u_{\min}(E)) = G(y_1(E)u_{\max}(E)) = E$.

Let us now prove (34). Note first that $H(y, E) \leq H(y, 0)$ for all y in $[\bar{y}(E), 1]$ and E in $(E_0, 0)$, and thus we have that for all E in $(E_0, 0)$,

$$\begin{aligned} \int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy &\leq -\frac{2E}{u_{\max}^2(E)} \int_{\bar{y}(E)}^1 \frac{1 - y^{-(p+1)}}{H^{3/2}(y, E)} dy \\ &\leq -\frac{2E}{u_{\max}^2(E)} \left(\frac{1 - \bar{y}(E)}{H(\bar{y}(E), E)} \right)^{3/2} \\ &\quad \times \int_{\bar{y}(E)}^1 \frac{1 - y^{-(p+1)}}{1 - y} \frac{dy}{\sqrt{1 - y}}, \end{aligned}$$

where we used the strict concavity of $H(y, E)$ with respect to y in $[\bar{y}(E), 1]$ for each E in $(E_0, 0)$, which implies that

$$H(y, E) \geq \frac{H(\bar{y}(E), E)}{1 - \bar{y}(E)} (1 - y).$$

Letting $M > 0$ be a bound for the function $y \rightarrow (1 - y^{-(p+1)})/(1 - y)$ on the interval $[0, 1]$ (recall that L'Hôpital's rule shows that this function has limit equal to $-(p+1)$ as $y \rightarrow 1-$), it follows that

$$\begin{aligned} \int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy &\leq -\frac{4E}{u_{\max}^2(E)} \left(\frac{1 - \bar{y}(E)}{H(\bar{y}(E), E)} \right)^{3/2} M \sqrt{1 - \bar{y}(E)} \\ &\rightarrow 0 \text{ as } E \rightarrow 0-, \end{aligned}$$

since $u_{\max}(E) \rightarrow \bar{u} > 0$, $\bar{y}(E) \rightarrow \bar{y}(0) > 0$, and $H(\bar{y}(E), E) \rightarrow H(\bar{y}(0), 0) > 0$ as $E \rightarrow 0-$. This shows that (34) holds and concludes the proof. $\square$

We now turn to the proof for the case that $p \in (-3, -2)$. This will require some new arguments when compared to the proofs of Lemmas 9 and 10. The main problem occurs because the function $H(y, E)$ is no longer concave in the y -variable when E is close to zero, which in turn prohibits comparisons between the graph of $H(y, E)$ and the various secant lines that were key to the proofs of Lemmas 9 and 10.

LEMMA 11. Assume that the conditions and notation of Lemma 8 hold. If $p \in (-3, -2)$, then

$$(35) \quad \lim_{E \rightarrow 0-} \Theta(E) = \lim_{E \rightarrow 0-} \int_{y_1(E)}^1 \frac{dy}{\sqrt{H(y, E)}} = \frac{\pi}{p+3}.$$

Proof. Again we start by examining the properties of the function $H(y, E)$. For each E in $(E_0, 0)$, $H(y, E)$ is positive for $y \in (y_1(E), 1)$ and zero at the endpoints $y = y_1(E)$ and $y = 1$. In what follows, y -subscripts will always denote partial derivatives with respect to the variable y . For all $E \in (E_0, 0)$ and all $y \in [y_1(E), 1]$, we have that

$$\begin{aligned} H(y, E) &= \frac{2E}{u_{\max}^2(E)} \left(1 - y^{-(p+1)} \right) + y^{-(p+1)} - y^2, \\ H_y(y, E) &= -(p+1) \left(\frac{-2E}{u_{\max}^2(E)} + 1 \right) y^{-(p+2)} - 2y \\ &= y \left(-2 - (p+1) \left(\frac{-2E}{u_{\max}^2(E)} + 1 \right) y^{-(p+3)} \right), \\ H_{yy}(y, E) &= (p+1)(p+2) \left(\frac{-2E}{u_{\max}^2(E)} + 1 \right) y^{-(p+3)} - 2. \end{aligned}$$

For each E in $(E^*, 0)$, $H(y, E)$ has a unique maximum in $[y_1(E), 1]$ at $y = \bar{y}(E)$, where

$$\bar{y}(E) = \left(\frac{-(p+1) \left(\frac{-2E}{u_{\max}^2(E)} + 1 \right)}{2} \right)^{\frac{1}{p+3}},$$

and a unique inflection point at $y = y^*(E)$, where

$$y^*(E) = \left(\frac{(p+1)(p+2) \left(\frac{-2E}{u_{\max}^2(E)} + 1 \right)}{2} \right)^{\frac{1}{p+3}}.$$

The inflection point $y^*(E)$ only exists in $[y_1(E), 1]$ provided that $y_1(E) < y^*(E)$, but since $y_1(E) \rightarrow 0$ as $E \rightarrow 0^-$, whereas $y^*(E)$ has a positive limit as $E \rightarrow 0^-$, this will happen for all $E < 0$ sufficiently close to 0. Therefore, we restrict the subsequent discussion to values of E sufficiently close to 0 in order for the inflection point $y^*(E)$ to exist in $[y_1(E), 1]$.

Similarly, we can define a function $H(y, 0)$ for $y \in [0, 1]$ as below and calculate its derivatives, which are denoted by primes $'$:

$$\begin{aligned} H(y, 0) &= y^{-(p+1)} - y^2 = y^2(y^{-(p+3)} - 1), \\ H'(y, 0) &= -(p+1)y^{-(p+2)} - 2y = y \left(-2 - (p+1)y^{-(p+3)} \right), \\ H''(y, 0) &= (p+1)(p+2)y^{-(p+3)} - 2. \end{aligned}$$

$H(y, 0)$ has two zeros, one at $y = 0$ and one at $y = 1$, has a unique maximum at $y = \bar{y}(0)$, where

$$\bar{y}(0) = \left(\frac{-(p+1)}{2} \right)^{\frac{1}{p+3}} = \lim_{E \rightarrow 0^-} \bar{y}(E),$$

and a unique inflection point at $y = y^*(0)$, where

$$y^*(0) = \left(\frac{(p+1)(p+2)}{2} \right)^{\frac{1}{p+3}} = \lim_{E \rightarrow 0^-} y^*(E).$$

Again, we shall assume that $E < 0$ is sufficiently close to zero to ensure that $y^*(0)$ belongs to the interval $[y_1(E), 1]$. Also note that for all $E \in (E^*, 0)$ and all $y \in$

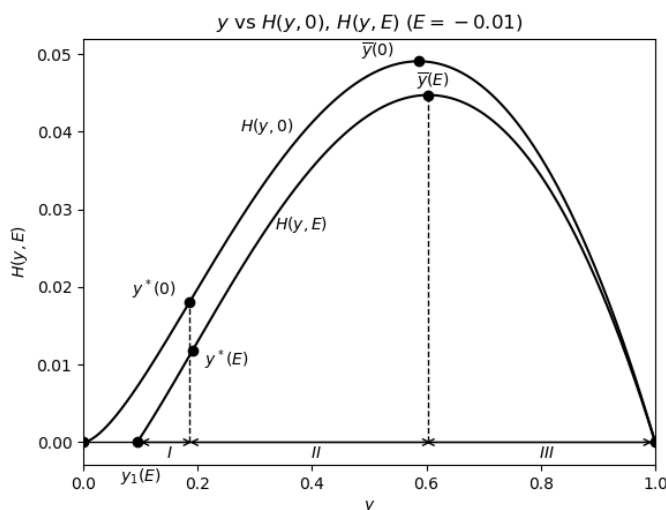


Fig. 3 Graphs of $H(y, E)$ and $H(y, 0)$ when $-3 < p < -2$.

$[y_1(E), 1]$,

$$(36) \quad H(y, E) \leq H(y, 0),$$

and that for all $E \in (E^*, 0)$ sufficiently close to zero,

$$y^*(0) < y^*(E) \text{ and } \bar{y}(0) < \bar{y}(E).$$

The graphs of the functions $H(y, E)$ and $H(y, 0)$ are illustrated in Figure 3.

By (29) and using the definition of $H(y, 0)$, we see that for all $E \in (E_0, 0)$,

$$\left| \Theta(E) - \frac{\pi}{p+3} \right| \leq \left| \int_{y_1(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy \right| + \int_0^{y_1(E)} \frac{dy}{\sqrt{H(0, y)}}.$$

As in the proof of Lemma 10, the second term on the right-hand side can be evaluated using the substitution $x = y^{(p+3)/2}$ and shown to converge to zero as $E \rightarrow 0^-$. Hence, it remains to show that the first integral converges to zero as $E \rightarrow 0^-$. To achieve this, it suffices, in view of (36), to show that

$$(37) \quad \lim_{E \rightarrow 0^-} \int_{y_1(E)}^{y^*(0)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy = 0,$$

$$(38) \quad \lim_{E \rightarrow 0^-} \int_{y^*(0)}^{\bar{y}(E)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy = 0,$$

$$(39) \quad \lim_{E \rightarrow 0^-} \int_{\bar{y}(E)}^1 \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy = 0.$$

Let's start by proving (37). It holds that

$$\begin{aligned} \int_{y_1(E)}^{y^*(0)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy &\leq \frac{-2E}{u_{\max}^2(E)} \int_{y_1(E)}^{y^*(0)} \frac{1 - y^{2-c}}{\sqrt{H(y, E)H(y, 0)}} dy \\ &\leq \frac{-2E}{u_{\max}^2(E)} \int_{y_1(E)}^{y^*(0)} \frac{1}{\sqrt{H(y, E)H(y, 0)}} dy. \end{aligned}$$

Note that, for each $E \in (E_0, 0)$, both $H(y, E)$ and $H(y, 0)$ are convex functions for $y \in [y_1(E), y^*(0)]$, and hence their graphs in these intervals are not below the tangent lines at the point on the respective graphs where $y = y_1(E)$:

$$\begin{aligned} H(y, E) &\geq H_y(y_1(E), E)(y - y_1(E)) \text{ for all } y \text{ in } [y_1(E), y^*(0)], \\ H(y, 0) &\geq H'(y_1(E), 0)(y - y_1(E)) + H(y_1(E), 0) \text{ for all } y \text{ in } [y_1(E), y^*(0)]. \end{aligned}$$

Consequently,

$$\begin{aligned} \int_{y_1(E)}^{y^*(0)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy &\leq \frac{-2E}{u_{\max}^2(E)} \frac{1}{\sqrt{H_y(y_1(E), E)}} \frac{1}{H'(y_1(E), 0)} \\ &\quad \int_{y_1(E)}^{y^*(0)} \frac{dy}{\sqrt{y - y_1(E)} \left(y - y_1(E) + \frac{H(y_1(E), 0)}{H'(y_1(E), 0)} \right)}. \end{aligned}$$

The latter integral is easily evaluated using the substitution $x = \sqrt{y - y_1(E)}$ and yields that

$$\begin{aligned} &\int_{y_1(E)}^{y^*(0)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy \\ &\leq \frac{-2E}{u_{\max}^2(E)} \frac{1}{\sqrt{H_y(y_1(E), E)}} \frac{1}{H'(y_1(E), 0)} \frac{2}{\sqrt{\frac{H(y_1(E), 0)}{H'(y_1(E), 0)}}} \\ &\quad \times \arctan \left(\sqrt{\frac{y^*(0) - y_1(E)}{H(y_1(E), 0)/H'(y_1(E), 0)}} \right) \\ &\leq \frac{-2E}{u_{\max}^2(E)} \frac{1}{\sqrt{H_y(y_1(E), E)}} \frac{1}{\sqrt{H'(y_1(E), 0)}} \frac{1}{\sqrt{H(y_1(E), 0)}} \pi. \end{aligned}$$

Using the explicit expressions for $H_y(y, E)$, $H(y, 0)$, and $H'(y, 0)$ given previously, and factoring appropriate powers of $y_1(E)$ from the three square roots, it follows that

$$\begin{aligned} &\int_{y_1(E)}^{y^*(0)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy \\ &\leq \frac{-2E}{u_{\max}^2(E)} \frac{1}{y_1^{\frac{-(p+2)}{2}}(E) \sqrt{-(p+1) \left(\frac{-2E}{u_{\max}^2(E)} + 1 \right) - 2y_1^{p+3}(E)}} \\ &\quad \times \frac{1}{y_1^{\frac{-(p+2)}{2}}(E) \sqrt{-(p+1) - 2y_1^{p+3}(E)}} \frac{1}{y_1^{\frac{-(p+1)}{2}}(E) \sqrt{1 - y_1^{p+3}(E)}} \pi \end{aligned}$$

$$(40) \quad = \frac{-E}{y_1^{\frac{-(3p+5)}{2}}(E)} \frac{2\pi}{u_{\max}^2(E)} \frac{1}{\sqrt{-(p+1) \left(\frac{-2E}{u_{\max}^2(E)} + 1 \right) - 2y_1^{p+3}(E)}} \\ \times \frac{1}{\sqrt{-(p+1) - 2y_1^{p+3}(E)}} \frac{1}{\sqrt{1 - y_1^{p+3}(E)}}.$$

Recall that $G(u_{\min}(E)) = E$ for all E in $(E_0, 0)$ or, more explicitly, that

$$\frac{1}{2}u_{\min}^2(E) + \frac{km}{l_0^2(p+1)}u_{\min}^{-(p+1)}(E) = E.$$

Dividing both sides by $y_1^{\frac{-(3p+5)}{2}}(E)$ and using that $y_1(E) = u_{\min}(E)/u_{\max}(E)$, we find that

$$\frac{E}{y_1^{\frac{-(3p+5)}{2}}(E)} = u_{\max}^{\frac{-(3p+5)}{2}}(E) \left(\frac{1}{2}u_{\min}^{\frac{3(p+3)}{2}}(E) + \frac{km}{l_0^2(p+1)}u_{\min}^{\frac{p+3}{2}}(E) \right) \rightarrow 0$$

as $E \rightarrow 0^-$; here we used that $p \in (-3, -2)$, $u_{\min}(E) \rightarrow 0$, and $u_{\max}(E) \rightarrow \bar{u} > 0$ as $E \rightarrow 0^-$. Using this fact, and taking the limit as $E \rightarrow 0^-$ in (40), shows that (37) holds.

Next we turn to a proof of (38). Before we begin, we point out an additional property of the function $H(y, E)$, namely, that

$$(41) \quad \frac{\partial H}{\partial E}(y, E) \geq 0$$

for all $E \in (E_0, 0)$ and all $y \in [y_1(E), 1]$. This is most easily seen from the definition of

$$H(y, E) = \frac{2E}{u_{\max}^2(E)} \left(1 - y^{-(p+1)} \right) + y^{-(p+1)} - y^2$$

and the fact that $u_{\max}(E)$ is increasing in E .

To establish (38), we can split the integral into two parts and then prove that

$$(42) \quad \lim_{E \rightarrow 0^-} \int_{y^*(0)}^{\bar{y}(0)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy = 0$$

and

$$(43) \quad \lim_{E \rightarrow 0^-} \int_{\bar{y}(0)}^{\bar{y}(E)} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} dy = 0.$$

We begin with a proof of (43). Recall first that for each E in $(E_0, 0)$, the function $y \rightarrow H(y, E)$ is increasing on the interval $[\bar{y}(0), \bar{y}(E)]$, and thus $H(\bar{y}(0), E) \leq H(y, E)$ in this interval. Then (41) and (36) imply that for all E in $(E_0/2, 0)$ and for all y in $[\bar{y}(0), \bar{y}(E)]$,

$$(44) \quad H(\bar{y}(0), E_0/2) \leq H(\bar{y}(0), E) \leq H(y, E) \leq H(y, 0).$$

It follows that the integrand in (43) is bounded by $2/\sqrt{H(\bar{y}(0), E_0/2)}$ when $E \in (E_0/2, 0)$. Recalling that $\bar{y}(E) \rightarrow \bar{y}(0)$ as $E \rightarrow 0^-$ then establishes (43).

To prove (42), it suffices to show that the integrand in (42) converges to zero, uniformly on $[y^*(0), \bar{y}(0)]$ as $E \rightarrow 0-$. Recall that for all E in $(E_0, 0)$, the map $y \rightarrow H(y, E)$ is increasing in $[y^*(0), \bar{y}(0)]$. Thus, a similar argument as the one used to establish (44) reveals that for all E in $(E_0/2, 0)$ and for all y in $[y^*(0), \bar{y}(0)]$,

$$(45) \quad H(y^*(0), E_0/2) \leq H(y^*(0), E) \leq H(y, E) \leq H(y, 0).$$

Consequently, we have that when $E \in (E_0/2, 0)$ and $y \in [y^*(0), \bar{y}(0)]$, then

$$\begin{aligned} \frac{1}{\sqrt{H(y, E)}} - \frac{1}{\sqrt{H(y, 0)}} &= \frac{H(y, 0) - H(y, E)}{\sqrt{H(y, E)}H(y, 0)} \\ &= \frac{-2E}{u_{\max}^2(E)} \frac{1 - y^{-(p+1)}}{\sqrt{H(y, E)}H(y, 0)} \\ &\leq \frac{-2E}{u_{\max}^2(E)} \frac{1}{\sqrt{H(y, E)}H(y, 0)} \\ &\leq \frac{-2E}{u_{\max}^2(E)} \frac{1}{H^{3/2}(y^*(0), E_0/2)}. \end{aligned}$$

Since the last bound is independent of y , and since $u_{\max}(E) \rightarrow \bar{u} > 0$ as $E \rightarrow 0-$, it follows that the integrand in (42) converges uniformly to zero on $[y^*(0), \bar{y}(0)]$ as $E \rightarrow 0-$, as desired. This concludes the proof of (42).

Finally, (39) can be established using the same arguments leading to (34) in the proof of Lemma 10. This part of the proof uses only the strict concavity of $H(y, E)$ in the interval $[\bar{y}(E), 1]$, and it remains valid in the present case where $p \in (-3, -2)$. $\square$

4. Bertrand's Theorem. We are ready to state and prove the main result of this paper.

THEOREM 1. *Assume that $h(r)$ is such that (SCO') holds. If all the solutions $\mathbf{r}(t)$ of (4) with bounded orbits have closed orbits, then necessarily $h(r) = kr^p$, where $p = -2$ (Newtonian gravitation) or $p = 1$ (Hookean gravitation).*

Proof. Suppose that (SCO') holds. Then (4) has a solution with a circular orbit of radius $r_0 = 1/u_0$ and uniform angular speed; see Remark 2 of section 3.1. Moreover, since this solution is Lyapunov stable, (4) has bounded solutions nearby. To the solutions $(u(\theta), v(\theta))$ of system (11)–(12) that correspond to these bounded solutions, we can associate the map $E \rightarrow \Theta(E)$ defined in (14) for all $E > E_0$ sufficiently close to E_0 . If these bounded solutions of (4) have closed orbits, then it follows from Lemma 5 that the map $\Theta(E)$ must be constant and this constant must be equal to a rational multiple of π . By continuity of the map $E \rightarrow \Theta(E)$, the value of this constant must equal $\Theta_0 := \lim_{E \rightarrow E_0+} \Theta(E)$, and by Lemma 6 the value of this limit is $\pi/\sqrt{G''(u_0)}$. But this limit must then be independent of u_0 , and Lemma 7 then implies that $h(r)$ must belong to $\mathcal{C}_{p>-3}$.

For $h(r)$ in $\mathcal{C}_{p>-3}$, the domain of the map $E \rightarrow \Theta(E)$ extends to all E in $(E_0, 0)$ if $p \in (-3, -1)$, and to all E in $(E_0, +\infty)$ if $p \geq -1$; see Lemma 8. Recall also from the remark in section 3.3 that when $p \in (-3, -1)$, the only solutions of (4) that are bounded correspond to solutions of system (11)–(12) that have orbits L_E , where E belongs exactly to this extended domain; when $p \geq -1$, then all solutions of (4) are bounded. Therefore, it suffices to show that the map $E \rightarrow \Theta(E)$ is constant on this extended domain. For this to happen, it must certainly hold that

$$(46) \quad \Theta_0 = \begin{cases} \lim_{E \rightarrow 0-} \Theta(E) & \text{if } p \in (-3, -1), \\ \lim_{E \rightarrow +\infty} \Theta(E) & \text{if } p \geq -1. \end{cases}$$

Recall from Remark 4 in section 3.1 that if $h(r)$ belongs to $\mathcal{C}_{p>-3}$, then $G''(u_0) = p+3$; consequently, $\Theta_0 = \pi/\sqrt{p+3}$.

On the other hand, the two limits in (46) equal $\pi/(p+3)$ if $p \in (-3, 1)$ (by Lemmas 10 and 11) and $\pi/2$ if $p \geq -1$, respectively (by Lemma 9). Solving the equation

$$\frac{\pi}{\sqrt{p+3}} = \frac{\pi}{p+3} \text{ with } p \in (-1, -3)$$

yields $p = -2$, which implies that $h(r) = kr^{-2}$ for some $k > 0$ (Newtonian gravitation). Solving the equation

$$\frac{\pi}{\sqrt{p+3}} = \frac{\pi}{2} \text{ with } p \geq -1$$

yields $p = 1$, which implies that $h(r) = kr$ for some $k > 0$ (Hookean gravitation). To conclude, we need to verify that for these two specific power laws corresponding to $p = -2$ and $p = 1$, the map $E \rightarrow \Theta(E)$ is indeed constant on its domain. But this turns out to be a standard exercises in calculus, as follows.

For $p = -2$, $G(u) = \frac{1}{2}u^2 - \kappa u$, where $\kappa := km/l_0^2$ is positive. The solutions $u_{\min}(E)$ and $u_{\max}(E)$ to the equation $G(u) = E$ can be calculated explicitly by completing the square, and then for every E in $(E_0, 0)$,

$$\begin{aligned} \Theta(E) &= \int_{u_{\min}(E)}^{u_{\max}(E)} \frac{du}{\sqrt{2(E - G(u))}} \\ &= \int_{\kappa - \sqrt{\kappa^2 + 2E}}^{\kappa + \sqrt{\kappa^2 + 2E}} \frac{du}{\sqrt{(\kappa^2 + 2E) - (u - \kappa)^2}} \\ &= \int_{-1}^1 \frac{dx}{\sqrt{1 - x^2}} = \pi. \end{aligned}$$

Here, we have used the substitution $x = \frac{u - \kappa}{\sqrt{\kappa^2 + 2E}}$.

As for $p = 1$, $G(u) = \frac{1}{2}u^2 + \frac{\kappa}{2}u^2$, where $\kappa = km/l_0^2$ is positive. Then $u_{\min}(E)$ and $u_{\max}(E)$ can be calculated by completing squares in the equation $G(u) = E$ again, and it follows that for all E in $(E_0, +\infty)$,

$$\begin{aligned} \Theta(E) &= \int_{u_{\min}(E)}^{u_{\max}(E)} \frac{du}{\sqrt{2(E - G(u))}} \\ &= \int_{\sqrt{E - \sqrt{E^2 - \kappa}}}^{\sqrt{E + \sqrt{E^2 - \kappa}}} \frac{udu}{\sqrt{(E^2 - \kappa) - (u^2 - E)^2}} \\ &= \frac{1}{2} \int_{-1}^1 \frac{dx}{\sqrt{1 - x^2}} = \frac{\pi}{2}. \end{aligned}$$

Here, we have used the substitution $x = \frac{u^2 - E}{\sqrt{E^2 - \kappa}}$. □

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